

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Forest resources & deforestation - Set 1

Mark scheme

AQA spec 3.5.3; Biological resources; textbook pp. 360-369

Marking guidance

- Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
- The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
- **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	1 mark each 1 mark for each of the 5 points below. 1 Timber: building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) 1 Fuel: fuelwood and charcoal - still the main energy source for most people in LEDCs 1 Food: fruit, nuts, bushmeat, livestock fodder – crops of forest origin (coffee, cacao, bananas, Brazil nuts) – pigs and chickens were forest animals 1 Fibres: paper (wood pulp cellulose), cotton, viscose / rayon 1 Medicines: quinine (cinchona), taxol (yew), aspirin (willow) – most species not yet researched Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	5

Question	Marking guidance	Marks
01.2	1 mark each 1 mark for each of the 3 points below. 1 Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time – total forest carbon (vegetation plus soil) is about double that in the atmosphere 1 Growing trees sequester CO₂ by photosynthesis and release oxygen – forest soils hold more carbon than the trees (especially boreal) 1 Planting / protecting forests removes CO₂ and reduces the enhanced greenhouse effect Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. residence time → the time a molecule stays in the atmosphere / atmospheric lifetime enhanced greenhouse effect → extra warming caused by added greenhouse gases Do not credit 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
01.3	1 mark each 1 mark for each of the 2 points below. 1 Highest transpiration of any ecosystem increases downwind rainfall – interception increases evaporation – deep organic soils retain water and delay runoff so river flow is more even and flooding reduced 1 Low albedo absorbs sunlight – heat stored and released at night reduces day-night temperature extremes – canopy microclimate: lower light, wind, higher humidity Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. transpiration → water lost as vapour from the leaves albedo → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back Do not credit 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	

Question	Marking guidance	Marks
02.1	Marks shared Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. up to 2 Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance – indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met up to 2 Single species: easy planting, thinning and felling over large areas – but each tree species supports its own community so few species overall up to 2 Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them up to 2 Simple age structure: uniform size makes management easy but gives little variety of light, temperature, wind – no old trees or dead wood for invertebrates and woodpeckers up to 2 Mixed woodland: varied niches, stratification, dead wood, indigenous trees each with their own dependent species Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • biodiversity → the variety of life: habitat, species and genetic diversity	6
02.2	1 mark each 1 mark for each of the 2 points below. 1 Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline 1 Continuing to grow them ties up land and capital for little extra timber – natural forests keep old and dead trees Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	1 mark each 1 mark for each of the 2 points below. 1 Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp 1 Wood fuel replaced by fossil fuels / electricity – so traditional management declined and woods became uniform Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	

Question	Marking guidance	Marks
03.1	1 mark each 1 mark for each of the 3 points below. 1 South America: high loss (5.1-5.5 Mha yr⁻¹) falling by about half by 2020 1 Africa: loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest 1 Europe: negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	Marks shared Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. up to 2 South America: clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export - HEP reservoirs - roads improving access - mining up to 2 Africa: growing rural populations clearing land for subsistence farming - fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy) - fodder cutting in dry areas up to 2 Both: hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted - urban expansion Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. - HEP → electricity generated from falling water Do not credit - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
03.3	1 mark each 1 mark for each of the 2 points below. 1 They create barriers fragmenting wildlife populations 1 Improved access makes it easier to remove timber and farm produce, so clearance spreads along them Also accept Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	1 mark for the point below. 1 Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades - slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	

Question	Marking guidance	Marks
04.1	1 mark each 1 mark for each of the 4 points below. 1 Less dead organic matter / leaf litter so less humus and slower soil formation – fewer nutrients returned 1 Loss of canopy and litter exposes soil to rain splash and wind 1 Decomposing roots no longer bind soil so erosion, gully and landslides increase (especially on slopes) 1 Reduced organic matter and biota lower infiltration and water retention Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. humus → dark, decomposed organic matter in soil infiltration → water soaking into the soil surface Do not credit 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	4
04.2	1 mark each 1 mark for each of the 4 points below. 1 Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind 1 Increased runoff: more flooding after rain, lower flows in dry periods – sedimentation of rivers 1 Increased albedo of cleared land (reflects more sunlight) 1 Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. transpiration → water lost as vapour from the leaves albedo → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back Do not credit 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss - a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
04.3	Any 2 of 3 1 mark each for any 2 of the 3 points below, in any order, maximum 2. 1 Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects 1 Loss of inter-species relationships (pollinators, seed dispersers) causing further declines 1 Loss of genetic resources / potential medicines before they are discovered Also accept - Any other correct, relevant point of the same kind - wording never has to match these notes. Do not credit - Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
05.1	1 mark each 1 mark for each of the 5 points below. 1 Harvest rate below the maximum sustainable yield – only economically valuable species (mahogany) selectively logged, not clear-felled 1 Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing – large seed trees and small trees left to grow 1 Buffer zones with no harvesting or machinery near rivers / wetlands 1 Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees 1 Each felled tree tagged and tracked to the retailer for FSC accreditation – rangers control illegal logging – tree nurseries replant over-exploited and ecologically important species Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • sustainable → meeting present needs without stopping future generations meeting theirs • maximum sustainable yield → the largest catch that can be taken without reducing the stock Do not credit • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
05.2	1 mark each 1 mark for each of the 3 points below. 1 The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it 1 Employment as rangers, forestry and tourism workers – ecotourists use local restaurants, produce and souvenirs – offcuts given to local carvers 1 Benefits give the community an incentive to protect the forest – carbon sequestration payments from companies fund management Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	1 mark each 1 mark for each of the 2 points below. 1 Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept – timber resource used over a longer period 1 Disadvantage: more labour-intensive and expensive – more tracks needed per tree Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	
06.1	Correct answer only 1 mark for the correct option. No mark if more than one box is ticked. 1 B	1
06.2	Any 4 of 6 1 mark each for any 4 of the 6 points below, in any order, maximum 4. 1 Harvesting no faster than the MSY - resource never depleted 1 Mixed species plantations - more food choices, relationships and niches 1 Indigenous species - support indigenous insects, birds 1 Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield 1 Selective logging - small clearings, less disturbance, long-term timber benefit 1 Retention of dead wood, buffer zones, corridors Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • biodiversity → the variety of life: habitat, species and genetic diversity • MSY → the largest catch that can be taken without reducing the stock Do not credit • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	